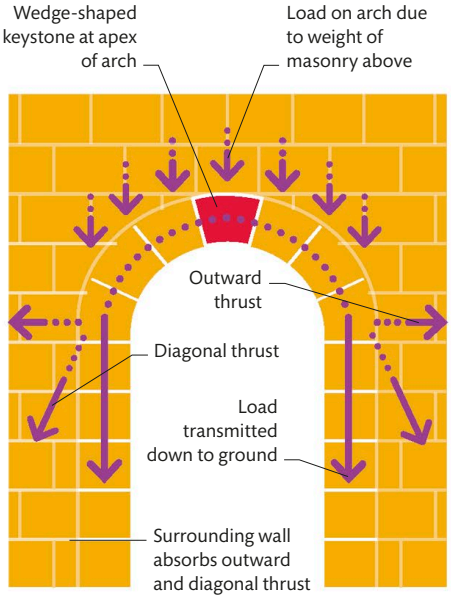
Arches and domes

For many traditionally constructed buildings, arches and domes are commonly used to span openings and big spaces, because they enable a large area to be covered with the least amount of supporting structure.

Arches

The simplest way to create an opening in a wall is to use two pillars (also called posts) with a flat beam across (the lintel) to carry the load above. However, this design is unable to support large loads and so does not allow for large openings. An arch can span wider openings, because the downward force from the weight of masonry pushes the individual blocks of the arch together, thereby utilizing the natural compressive strength of materials such as brick and stone. While an arch is under construction, it has to be supported by scaffolding until the keystone is in place to lock the structure securely.



Forces in an arch
The load on an arch is carried along the curve and down. The load also produces outward and diagonal thrusts, which are countered by the surrounding wall or by buttresses.

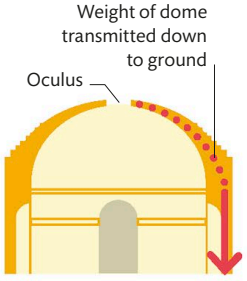
Domes

A dome is like an arch rotated in a circle to form a three-dimensional shape. Like an arch, a dome is self-supporting, with all the weight transferred down to the structure on which it rests. However, unlike an arch, a dome does not require a keystone to lock it in place, and domes are stable during construction as each level is a complete and self-supporting ring. The weight of the dome creates forces that thrust outward. To counter this outward force, tension rings, which act like hoops on a barrel, are wrapped around the dome.

THE **WORLD'S FIRST GEODESIC DOME** OPENED IN **1926** IN GERMANY. IT WAS **82 FT (25 M) IN DIAMETER**

ROME'S PANTHEON

Almost 2,000 years after it was built, the dome of the Pantheon is still the world's largest unreinforced concrete dome, with an inner diameter of about 142 ft (43.3 m) and a weight of 5,000 tons (4,535 metric tons). To minimize the dome's weight, the concrete is thinner at the top and thicker at the base. The weight is further reduced by indentations in the dome, called coffers, and by a 26 ft (8 m) diameter hole at the apex, known as an oculus.



Brunelleschi's dome
The dome of Florence cathedral, commonly known as Brunelleschi's dome after its designer, is the largest masonry dome ever built, at about 148 ft (45 m) across and rising to 376 ft (114.5 m) above ground level. It consists of two concentric octagonal domes, or shells: an inner shell visible from inside the cathedral and a larger external shell.